UrubutoPay

Integration Guide

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1. Introduction

This guide provides a step-by-step overview for integrating our payment gateway into your application or website. It details the required APIs, security protocols, and best practices to ensure seamless, secure, and reliable payment processing. By following this guide, developers can quickly enable key payment functionalities, including transactions, refunds, webhooks, bulk transfers, recurring payments, payment links, wallet, tipping, notifications and so on.

2. Integration Overview

This section lists the services used to connect your system with the third-party interface. It explains the APIs, webhooks, and other components involved to ensure smooth communication and successful transaction processing.

The following services enable communication with your system via a webhook URL.

2.1 Webhooks

2.1.1 Authentication

An API must be provided by your system to request and obtain a valid Bearer token. The token remains valid for at least 2 hours. This enables us to deliver secure webhook notifications and perform payer validation in a secure way.

2.1.2 Payer validation

A webhook endpoint URL should be provided by your platform, which we will use to validate the biller/payer against your system.

2.1.3 Payment Notification

A webhook URL provided by you, used for payment notifications. We will only send notifications when a payment is successful or pending settlement. Funds will be sent to your trust bank account registered with our partner banks (BK, SACCO, BOA, I&M).

2.1.4 Payment Callback

A webhook API URL provided by you, used for immediate callbacks after a payment is initiated, pending, failed, successful, or awaiting settlement.

2.1.5 Reversal Payment Notification

API provided by your system to handle canceled payment notifications and respond with success or failure

2.2 Payment Processing

UrubutoPay offers public APIs for payment collections. Merchants can use these APIs to initiate payments via wallet or card channels, verify transaction status, retrieve collected funds, and perform payer KYC validation through wallet channels (MTN MOMO & AIRTEL Money).

2.3 Bulk Transfers

UrubutoPay provides public APIs for bulk disbursements. Merchants can use these APIs to initiate bulk transfers via the **UrubutoPay wallet** to **MTN**, **MTN** to **MTN**, **AIRTEL MONEY** to **AIRTEL MONEY**, or **UrubutoPay wallet** to **AIRTEL MONEY**.

The UrubutoPay wallet is an integrated wallet that enables merchants to transfer funds to telecommunications wallets commonly used in Rwanda (MTN & AIRTEL Money). With those APIs can also be used to verify your wallet balance and other related informations.

Note: To use the Bulk Transfers APIs, you must provide the public IP addresses of your system for whitelisting, ensuring secure communication across platforms

2.4 Recurring Payment

UrubutoPay offers public APIs for managing recurring payments, enabling merchants to initiate, execute, renew, and cancel recurring transactions.

Merchants can process recurring payments through both telecommunication channels (MTN MOMO) and card channels (MASTERCARD & VISA) via UrubutoPay.

Note: To use the Recurring APIs, you must provide the public IP addresses of your system for whitelisting, ensuring secure communication across platforms

2.5 Payment links

Merchants can use the UrubutoPay merchant portal to create payment links for collecting payments from their clients

2.6 Connect

UrubutoPay Connect enables merchants to manage USSD sessions and communications (SMS and Email). With communications, merchants can be able to send bulk SMS to their clients.

These capabilities are accessible through public APIs as well as the UrubutoPay merchant portal.

2.7 BK Pay & E-Kash

Merchants using UrubutoPay are provided with a unique public code for collecting payments from clients. All collected funds are automatically deposited into the merchant's Bank of Kigali account.

By using the UrubutoPay merchant code through E-kash, you can collect payments from telecommunication wallets (**MTN & AIRTEL MONEY**) into UrubutoPay. All collected funds are then automatically deposited into the merchant's Bank of Kigali account

3. Webhooks

UrubutoPay leverages webhooks to keep merchants updated in real time. This ensures merchants are promptly notified of critical events, including payer validation, authentication, callbacks notification, payment notification, and reversals.

3.1 Authentication

This section explains the importance of authentication and how it enables secure communication between your platform and the third-party platform (UrubutoPay).

As mentioned in section 2.1, authentication is provided via an API URL that you share with us alongside the username and password we will use to authenticate in your system. This allows us to perform the following actions: validate the payer, send callbacks, notify your system of successful payments, and notify your system of canceled payments.

The following tables show basic properties of how the authentication endpoint will look like.

3.1.1 Request Method

HTTP request method	URI path segment
POST	/auth/authenticate

3.1.2 Request Header

Header name	Type
Content-Type	application/json

3.1.3 Request Payload

Field Name	Type	Description	Required
user_name	string	This username, which you must share with us, is required to authenticate into your platform	yes
password	string	This password, which you must share with us, is required to authenticate into your	yes

		platform	

3.1.4 Successful Response Object

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains token	yes
status	number	Http status code. Must be 200	yes
message	string	Any successful confirmation message.	no

3.1.4.1 Successful Response Data Object

Field Name	Type	Description	Required
token	string	Bearer JWT token. This token is valid for 24 hours	yes

3.1.5 Failure Response Object

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
status	number	Http status code. Any http code except 200, 201,200	yes
message	string	Failure reason	yes

Header's example:

Content-Type: application/json

Request Payload

```
{
"user_name": "bkTechPymtGtwy",
"password": "myPss@2020"
}
```


Success Response:


```
{
"timestamp": "2021-09-20 06:15:38",
"message": "Successful",
"status": 200,
"data": {
"token": "Bearer eyusdfsdtoken"
}
}
```

Failure Response:


```
{
"timestamp": "2021-09-22 23:55:05",
"message": "Invalid credentials",
"status": 401
}
```

3.2 Payer Validation

This section explains how payer validation works, showing the fields, you will receive from us and the expected response you should provide.

Payer validation is an API URL that you provide to the third-party platform (UrubutoPay), allowing UrubutoPay to verify the payer from your system before processing a payment.

The API request includes the **merchant code** and **payer code**. Upon receiving a request, you should check whether the payer or biller exists in your system.

- If the payer exists, respond with a **successful response object**.
- If the payer does not exist, respond with a **failure response object**.

3.2.1 Request Method

HTTP request method	URI path segment
POST	/payer/verify

3.2.2 Request Header

Header name	Type
Content-Type	application/json
Authorization	Bearer token. (JWT token or any token just make sure it expires in 24 hours)

Note: The token sent in the Authorization header is the one you provide us after successful authentication.

3.2.3 Request Payload

Field Name	Type	Description	Required
merchant_code	string	A unique merchant code generated by UrubutoPay, serving as a public identifier linked to your merchant account	yes
payer_code	string	The Bill ID is generated by your	yes

		system and is linked to your biller or payer	

3.2.4 Successful Response Object

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains token	yes
status	number	Http status code. Must be 200	yes
message	string	Any successful confirmation message.	no

3.2.4.1 Successful Data Response Object

Field Name	Type	Description	Required
merchant_code	string	Merchant code generated by UrubutoPay	yes
payer_code	string	Payer code generated by your syste	yes
payer_names	string	Name of the biller/payer	yes
amount	number	Amount that will be paid. It is optional and will only be applied when payer_must_pay_total_amount is YES	no
currency	string	RWF	no
payer_must_pay_total_amount	string	YES / NO	no
comment	string	Observation / payment comment	no

Note: Kindly note that upon successful payer validation, we will link the services associated with your merchant and other merchant configurations. Those fields that will be linked on the response are the following:

Field Name	Type	Description	Required

merchant_name	string	Names of the merchant	yes
merchant_short_code	string	Unique short code generated by UrubutoPay. This code allows our merchant to receive payment accross the country through **E-Kash/ Rswitch**	no
payer_names	string	Name of the biller/payer	yes
payer_to_be_charged	string	**YES / NO** – Merchant configuration indicating whether UrubutoPay will charge the payer directly or charge the merchant based on a commission rate.	yes
accept_card_payment	string	**YES/NO -** Merchant configuration indicating whether merchant accept card payment or not	yes
card_currency	string	**RWF / USD** – Currently, UrubutoPay supports these two currencies for card payments.	no
need_term_and_year	string	**YES / NO** – Setting that determines if the payer can choose the quota, term, or semester they are paying for. This is mainly applicable to school merchants	no
number_of_term	number	UrubutoPay allows merchants to define the number of terms that are acceptable during the payment process	no
term_label_name	string	UrubutoPay allows	no

		merchants to define the custom name of terms that will be displayed on checkout page	
commission_rate	number	Merchant rate applied. This is only applicable for WALLET payment channels, such as MTN or AIRTEL Money.	yes
card_commission_rate	number	Merchant rate applied. This is only applicable for card payments.	no
wallet_bank_settlement	string	The configured default bank used to transfer collected funds to the merchant's bank account. This applies only when the payer uses WALLET payment channels. Supported banks: **BK, SACCO, I&M, and BOA** **Note:** Services shown on our Hosted Checkout, UrubutoPay App, or USSD are limited to those linked to the default wallet bank settlement. Services associated with other banks will appear only via the respective partnered bank channels	yes
services	Array object	Services registered for the merchant. Each service includes a	yes

		name, service code, account number, and bank name	

3.2.4.2 Services Object

Field Name	Type	Description	Required
name	date	Name of the service	yes
service_code	number	Unique code generated by Urubutopay	yes
bank_name	string	Bank name prefix registred on the service.	yes
account_number	string	Bank account number	yes
service_id	number	Unique service id	yes
amount	number	Amount – This value is applicable only when payer_must_pay_total_amount is set to **YES**	no
is_recurring_enabled	boolean	Indicates whether the service is enabled for use with recurring payments	no

3.2.5 Failure Response Object

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
status	number	Http status code. Any http code except 200, 201,200	yes
message	string	Failure reason	yes

Header's example:

Content-Type: application/json

Authorization: Bearer token

Request Payload


```
{
"merchant_code": "TH456****",
"payer_code": "***********"
}
```


Success Response:


```
{
"status": 200,
"message": "validated successfully",
"timestamp": "2025-09-30 6:03:17 PM",
"data": {
"merchant_code": "TH15082215",
"payer_code": "201863794",
"payer_names": "ISHEJA KEZA LEILA",
"currency": "RWF",
"payer_must_pay_total_amount": "NO",
"amount": "500",
"comment": "school fees"
}
}
```

Failure Response:


```
{
"status": 404,
"message": "no data found for the given payer code",
"timestamp": "2025-09-30 6:04:10 PM"
}
```

3.3 Callback

The callback webhook allows UrubutoPay to notify merchants. It supports four types of callbacks: payment, bulk disbursement debit, bulk disbursement payment, and recurring.

The following are the prefixes for the callback types you will receive from us:

- **PAYMENT**: Sends a notification to the merchant whenever a payment status is updated. For example, when a payment changes from PENDING to PENDING_SETTLEMENT, from PENDING_SETTLEMENT to VALID, or from PENDING to FAILED, a webhook will be triggered.

- **BULK_DISBURSEMENT_DEBIT**: Notifies the merchant when funds are debited from their account for a bulk disbursement. The merchant account can be an UrubutoPay wallet, MOMO, AIRTEL, or BK Bank. Additionally, a webhook is triggered whenever a disbursement collection status changes, such as from PENDING to SUCCESSFUL or from PENDING to FAILED.

- **BULK_DISBURSEMENT_PAYMENT**: Notifies the merchant when a transfer in a bulk disbursement is completed, whether it is marked as **SUCCESSFUL** or **FAILED.**

- **RECURRING:** Notifies the merchant whenever a recurring payment is executed or its status changes

3.3.1 Payment callback

Payment callback payload will be sent based on the configuration set on your merchant account under UrubutoPay. Kindly note that by sending payment callback, we first authenticate using the authentication API URL you share with us. Then we use Bearer token shared and provide it in headers **Authorization.**

3.3.1 Request Payload when UrubutoPay Is Allowed to Perform Payer Validation

Field Name	Type	Description	Required
status	number	An HTTP status code. You will receive status 200	yes
transaction_status	string	UrubutoPay Transaction Status. The current transaction statuses are: **INITIATED,**	yes

		PENDING, PENDING_SETTLEMENT, VALID, FAILED, and **CANCELED**.	
merchant_code	string	Merchant Code Used for Payment Processing	yes
payer_code	string	Payer Code or Bill ID Generated by the Merchant Platform	yes
payment_channel	string	The payment channel can be one of the following: **WALLET, CARD, BANK,** or **AGENT**.	yes
payment_channel_name	string	The channel name can be one of the following: **MOMO, AIRTEL_MONEY, MASTERCARD, VISA, BK, IM, BOA NA,** or **SACCO**.	yes
amount	number	Amount paid	yes
currency	string	RWF or USD	yes
payment_date_time	date time	Transaction date	yes
callback_type	string	The callback type sent will be **PAYMENT**	yes
metadata	object	Additional details provided by the merchant.	no
slip_number	string	Bank Slip Number (for VALID transactions with instant settlement requested by the merchant)	no
transaction_id	string	Transaction ID generated by merchant platform	yes

3.3.2 Request Payload When UrubutoPay is not Allowed to Perform Payer Validation

Field Name	Type	Description	Required
status	number	An HTTP status code. You will receive status 200	yes
transaction_status	string	**UrubutoPay Transaction Status** The current transaction statuses are: **INITIATED, PENDING, PENDING, PENDING_SETTLEMENT, VALID, FAILED,** and **CANCELED**.	yes
merchant_code	string	Merchant Code Used for Payment Processing	yes
payer_code	string	Payer Code or Bill ID Generated by the Merchant Platform	yes
payment_channel	string	The payment channel can be one of the following: **WALLET, CARD, BANK,** or **AGENT**.	yes
payment_channel_name	string	The channel name can be one of the following: **MOMO, AIRTEL_MONEY, MASTERCARD, VISA, BK, IM, BOA NA,** or **SACCO**.	yes
amount	number	Amount paid	yes
currency	string	RWF or USD	yes
payment_date_time	date time	Transaction date	yes
callback_type	string	The callback type sent will be **PAYMENT**	yes
metadata	object	Additional details provided by the merchant.	no
slip_number	string	Bank Slip Number (for VALID transactions with instant settlement requested by the	no

		merchant)	
internal_transaction_id	string	Transaction ID generated by UrubutoPay	yes
transaction_id	string	Transaction ID generated by merchant platform	no
commission_amount	number	The amount charged as commission	yes

3.3.2 Success Response

We expect a response from you upon receiving the request. Receiving this response format allows us to notify the client of a successful payment via SMS or Email.

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains transaction details	yes
status	number	Http status code. Must be 200	yes
message	string	Any successful confirmation message.	no

3.3.2.1 Success Data Object Response

Field Name	Type	Description	Required
internal_transaction_id	string	This will be merchant transaction Id.	yes
external_transaction_id	string	UrubutoPay transaction ID.	yes
payer_phone_number	string	Payer phone number	yes
payer_email	string	Email of the payer	no

3.3.2 Failure Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
status	number	Http status code. Must be 200	yes
message	string	Any successful confirmation message.	no

Headers:
Content-Type: application/json
Authorization: Bearer eystoekene

Request payload (When Payer validation was performed initially):


```
{
"status": 200,
"transaction_status": "VALID",
"transaction_id": "33202509121119105806",
"merchant_code": "TH15082215",
"payer_code": "2018637405",
"payment_channel": "CARD",
"payment_channel_name": "MASTERCARD",
"amount": 10,
"currency": "RWF",
"payment_date_time": "2025-09-12 09:19:10",
"callback_type": "PAYMENT"
}
```

Success Response:


```
{
"status": 200,
"data": {
"internal_transaction_id": "563343434343434",
"external_transaction_id": "11546464564567567",
"payer_phone_number": "25078****************",
"payer_email": "gr@bk.rw"
}
}
```


Failure Response:


```
{
"message": "internal server error"
}
```

3.4 Notification

The notification webhook functions similarly to the callback, with one key difference: the notification webhook sends updates to the merchant twice. First, when the payment enters the pending settlement state, and again, when the payment is successful. In contrast, the callback notifies the merchant on every transaction status change.

Second, notification tells merchant if payment is in pending settlement or successful. Pending settlement, we don't provide bank slip number while on successful we provide bank slip number.

3.4.1 Method

HTTP request method	URI path segment
POST	https://yourplatform.com/payment/webhook/notification

3.4.2 Header

Header name	Type
Content-Type	application/json
Authorization	Bearer jwttoken

3.4.3 Parameters

Field Name	Type	Description	Required
merchant_code	string	Merchant Code Used for Payment Processing	yes
payer_code	string	Payer Code or Bill ID Generated by the Merchant Platform	yes
payment_channel	string	The payment channel can be one of the following: **WALLET, CARD, BANK,** or **AGENT**.	yes
payment_channel_name	string	The channel name can be one of the following: **MOMO, AIRTEL_MONEY, MASTERCARD, VISA, BK, IM, BOA NA,** or	yes

		SACCO.	
amount	number	Amount paid	yes
currency	string	RWF or USD	yes
payment_date_time	date time	Transaction date	yes
metadata	object	Additional details provided by the merchant.	no
slip_number	string	Bank Slip Number (for VALID transactions with instant settlement requested by the merchant)	no
transaction_id	string	Transaction ID generated by UrubutoPay	yes
payment_purpose	string	Name of service paid	yes
payment_purpose_code	string	Service code	yes
bank_name	string	Bank name. This is only applied for those merchants that need Realtime settlement	no
bank_account	string	Bank account number. This is only applied for those merchants that need Realtime settlement	no
observation	string	comment	yes
term	string	Only applied mostly on Schools. This could be a semester, etc.	no
academic_year	string	Applied on schools	no
initial_slip_number	string	Bank slip number — previously used in the v1 API. It has now been replaced with slip_number. This field applies only to merchants who require real-time settlement.	no

3.4.2 Success Response

We expect a response from you upon receiving the request. Receiving this response format allows us to notify the client of a successful payment via SMS or Email.

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains transaction details	yes
status	number	Http status code. Must be 200	yes
message	string	Any successful confirmation message.	no

3.4.2.1 Success Data DTO

Field Name	Type	Description	Required
internal_transaction_id	string	This will be merchant transaction Id.	yes
external_transaction_id	string	UrubutoPay transaction ID.	yes
payer_phone_number	string	Payer phone number	yes
payer_email	string	Email of the payer	no

3.4.2 Failure Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
status	number	Http status code. Must be 200	yes
message	string	Any successful confirmation message.	no

Headers:

Content-Type: application/json
Authorization: Bearer eystoekene

Request Payload:


```
{
"transaction_id": "1190********",
"merchant_code": "TH45454565",
"payer_code": "PYCD4533453553",
"academic_year": "2025",
"payment_channel": "WALLET",
"payment_channel_name": "MOMO",
"bank_name": "BK",
"bank_account": "45454",
"payment_purpose": "service paid",
"payment_purpose_code": "code of service paid",
"amount": "500",
"currency": "RWF",
"term": "1",
"observation": "Payment of Nov 2025",
"initial_slip_number": "FT45454544",
"slip_number": "FT45454544",
"payment_date_time": "2025-11-10 14:50:45"
}
```


Success Response:

```
{
"status": 200,
"data": {
"internal_transaction_id": "563343434343434",
"external_transaction_id": "11546464564567567",
"payer_phone_number": "25078****************",
"payer_email": "gr@bk.rw"
}
}
```


Failure Response:


```
{
"message": "internal server error"
}
```

4. Payment Collections API Guides

There are several ways merchants can collect payments:

Non-integrated merchants can accept payments through our hosted checkout page or by generating shareable payment links

Integrated merchants can collect payments using our public APIs or a hybrid approach:

- **Hybrid approach**: Merchants can collect payments in the following ways:
  - **Hosted Checkout Page**: Redirects the payer to our hosted checkout. Once the payment is successful, your system will be notified via the webhook API URL.
  - **USSD Platform**: Payer dials the USSD code (***775#**) and provides the merchant code and payer code to complete the payment.
  - **UrubutoPay App**: Payer can complete the payment directly using the UrubutoPay mobile app.
- **Public APIs**: Provide full customization, allowing you to build payment flows directly within your platform.

4.1 Hybrid Approach – Hosted Checkout Page

Payment can be initiated in two ways:

1. Through the merchant platform: The merchant redirects the payer to UrubutoPay using a redirection URL, for example:

https://urubutopay.rw/pay-now/initiate/?mhcd=TH56565&pycd=UG43534534

The payer is then taken to the hosted checkout page, where the payment is processed.

Success Page Response

Merchant Details

K

KIGALI PARENTS SCHOOL

MERCHANT CODE
TH15082215

PAYER CODE / BILL ID
201863794

Dear ISHEJA KEZA LEILA

Please provide the following details and proceed

Services *

Services

Amount (RWF) *

Amount

Semester

Select Semester

Choose a Payment Mode

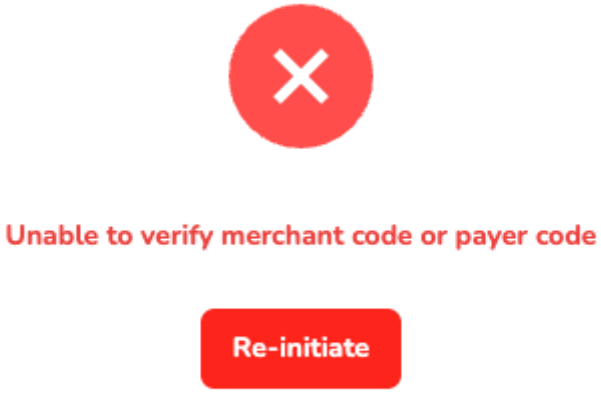
MoMo | airtel money | VISA

I have read and constent to UrubutoPay processing my information in accordance with the Privacy statements and Terms & conditions

I have read and accepted terms and condition of **KIGALI PARENTS SCHOOL**

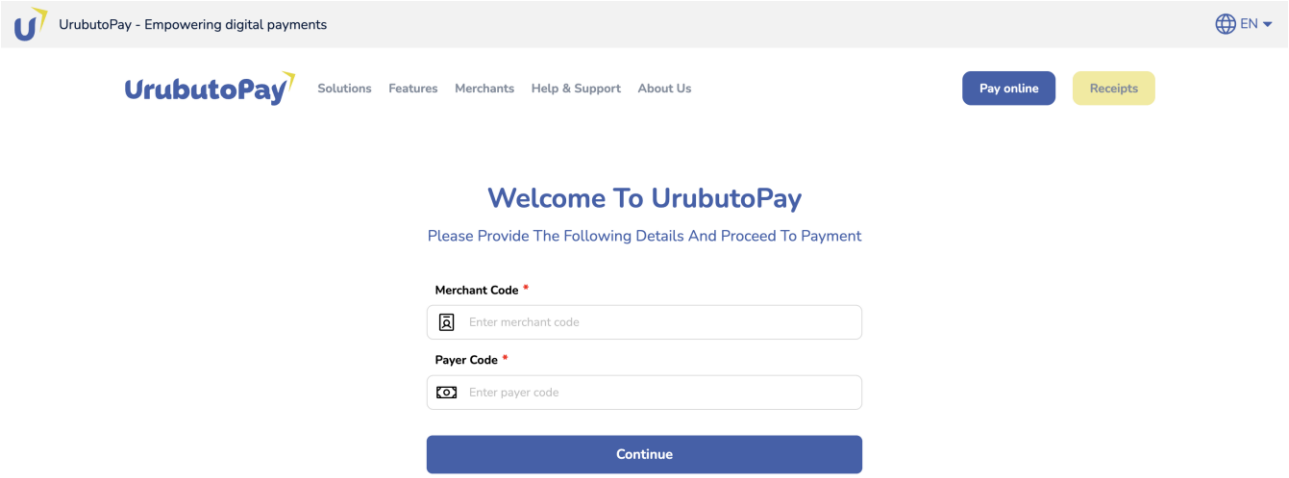
Pay Now

Failed Page Response



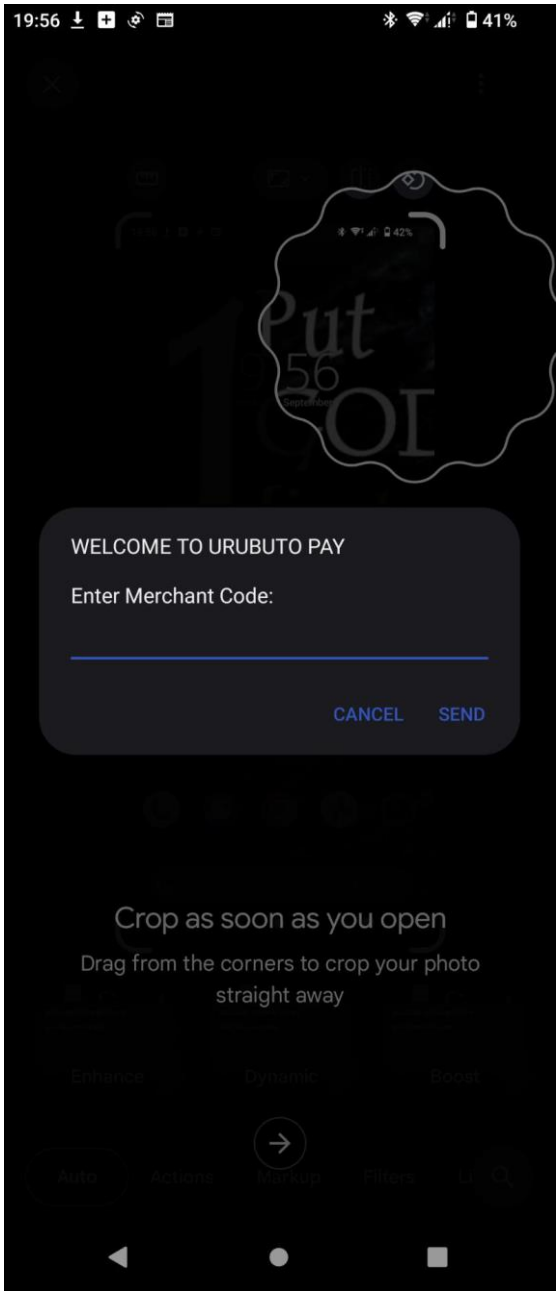
2. Through UrubutoPay: The payer visits the website, selects **"Pay Online"** from the header menu, enters the merchant and payer codes on the Pay-Now page, and clicks **Continue** to reach the checkout page.

https://urubutopay.rw/pay-now

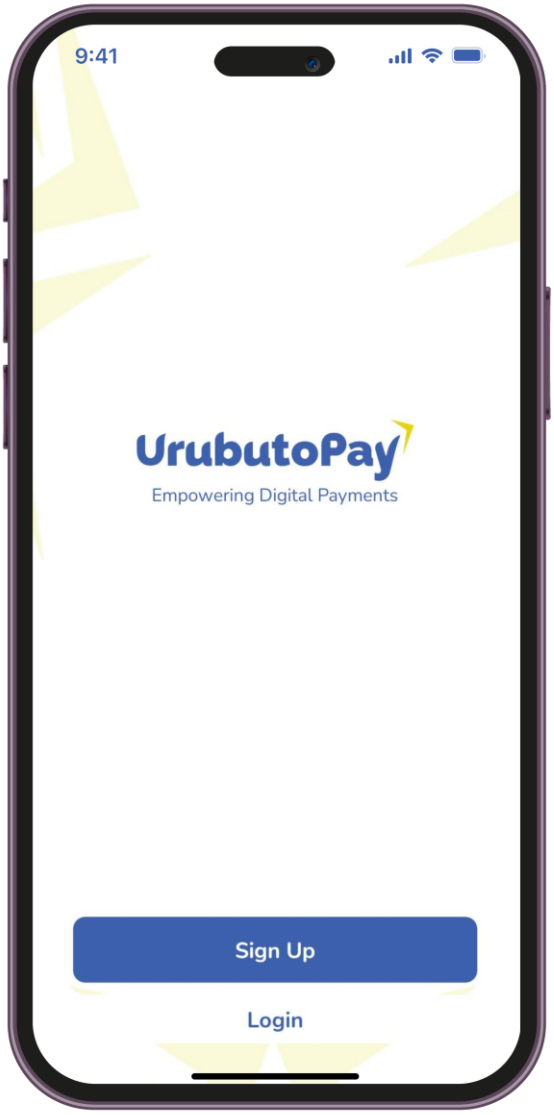


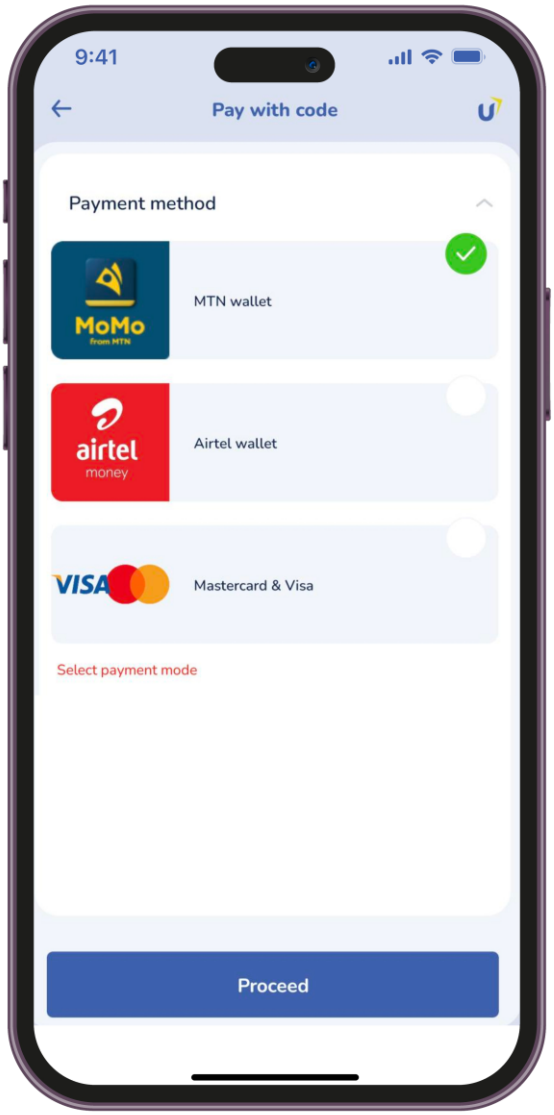
4.2 Hybrid Approach – USSD Platform

Customer dials the USSD code (***775#**) and provides the merchant code, and payer code then follows the process.



4.3 Hybrid Approach – UrubutoPay App




9:41
Pay with code
Payment method
MTN wallet
Airtel wallet
Mastercard & Visa
Select payment mode
Proceed

4.4 Public Payment collection APIs

Integrated merchants may leverage public APIs for payment processing, with complete flexibility to design their workflow as desired.

Note: To submit any API request, you must provide an API Key generated from the UrubutoPay portal under the developer menu. Payment requests won't be processed without this API Key. The key should be included in the request header as **Authorization**

4.4.1 Initiate Payment

An API designed to enable merchants to send payment initiation requests from their platform to UrubutoPay.

4.4.1.1 Request Method

HTTP request method	URI path segment
POST	/api/v2/payment/initiate

4.4.1.2 Request Header

Header name	Type
Content-Type	application/json
Authorization	api-key

4.4.1.3 Request Payload

Field Name	Type	Description	Required
merchant_code	string	A unique merchant code generated by UrubutoPay, serving as a public identifier linked to your merchant account	yes
payer_code	string	The Bill ID is generated by your system and is linked to your biller or payer	yes

payer_names	string	The names of the payer	yes
payer_email	string	The email of the payer	no
phone_number	string	Payer's phone number. **Note**: For wallet channels (**MOMO & AIRTEL MONEY**), only Rwandan phone numbers are accepted, whereas **CARD** payments allow international phone numbers	yes
amount	number	The Amount to be paid.	yes
channel_name	string	The available payment channels through the merchant platform are limited to **MOMO**, **AIRTEL_MONEY**, and **CARD**	yes
transaction_id	string	A unique transaction ID generated by the merchant's platform.	yes
service_code	string	Unique service code issued by UrubutoPay to identify the specific service a client has paid for.	yes
redirection_url	string	Applicable only for the CARD payment channel. The redirection URL is used to redirect the payer to the merchant's checkout page after the payment is successful or fails	yes
card_type_to_be_used	string	This applies exclusively to **CARD**	no

		payments using a BK PrePaid card.	

4.4.1.4 Successful Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains initiated payment details	yes
status	number	Http status code. Must be 200	yes
message	string	Successful confirmation message	Yes

4.4.1.5 Successful Data Object Response

Field Name	Type	Description	Required
payment_channel	string	Payment channels. **WALLET** or **CARD**	yes
payment_channel_name	string	**MOMO** or **AIRTEL_MONEY**	yes
transaction_status	string	UrubutoPay Transaction statuses. We have INITIATED, PENDING, FAILED, VALID, PENDING_SETTLEMENT	yes
internal_transaction_ref_number	string	Unique transaction id generated by UrubutoPay	Yes
external_transaction_ref_number	string	Unique transaction id generated by merchant platform	yes
merchant_code	string	Merchant code of merchant	yes
payer_code	string	The payer code or bill ID used.	yes
paid_mount	number	The amount paid	yes
payer_names	string	The names of the payer	yes
payer_phone_number	string	The phone number used for payment	yes
currency	string	RWF / USD	yes

4.4.1.6 Failure Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
status	number	Http status code. Any http code except 200, and 201	yes
message	string	Failure message	yes

Header's example:
Content-Type: application/json
Authorization: upg:test_sk_apikeywrwerwerwerwerwer

Wallet Request Payload:

```
{
"amount": 5,
"channel_name": "MOMO",
"merchant_code": "TH15082215",
"payer_code": "201863794",
"payer_names": "James Bond",
"phone_number": "25078*********",
"service_code": "stockings-1681",
"transaction_id": "45903423423432423432",
"payer_email": "janedoe@example.com"
}
```

CARD Request Payload:


```
{
"amount": 5,
"channel_name": "CARD",
"merchant_code": "TH15082215",
"payer_code": "201863794",
"payer_names": "James Bond",
"phone_number": "25078*********",
"service_code": "stockings-1681",
"transaction_id": "45903423423432423432",
"payer_email": "janedoe@example.com",
"redirection_url": "https://yourdomain/payment-confirmation/yourtransactionidoranything"
}
```


WALLET Successful Response:


```
{
"timestamp": "2025-10-01 07:42:46",
"status": 200,
"message": "Thank you for initiating a new Payment. Check your pending transactions on 182*7*1# to confirm.",
"data": {
"payment_channel": "WALLET",
"payment_channel_name": "MOMO",
```

"transaction_status": "INITIATED",

"internal_transaction_ref_number": "33202510010942449208",

"transaction_date": "2025-10-01T07:42:44.985Z",

"external_transaction_ref_number": "45903423423432423432",

"merchant_code": "TH15082215",

"payer_code": "201863794",

"paid_mount": 5,

"payer_names": "James Bond",

"payer_phone_number": "250786601003",

"currency": "RWF"

}

}

CARD successful response:

{

"timestamp": "2025-10-01 07:56:34",

"status": 200,

"message": "Thank you for initiating a new Payment. Continue to process Card Payment",

"card_processing_url": "https://staging.urubutopay.rw/pay-now/card/process/33202510010956326446",

"url_validity": "30 min",

"data": {

"payment_channel": "CARD",

"payment_channel_name": "NA",

```
"transaction_status": "INITIATED",
"internal_transaction_ref_number": "33202510010942449208",
"transaction_date": "2025-10-01T07:42:44.985Z",
"external_transaction_ref_number": "45903423423432423437",
"merchant_code": "TH15082215",
"payer_code": "201863794",
"paid_mount": 5,
"payer_names": "James Bond",
"payer_phone_number": "250786601003",
"currency": "RWF"
}
}
```


Note: After a successful card payment initiation, use the card_processing_url provided in the response to redirect the client to the card processing page, where they will enter their card details.

Failure Response:

```
{
"timestamp": "2025-10-01 07:59:39",
"status": 400,
"message": "redirection url should be valid url"
}
```

```
{
"status": 404,
"message": "no active merchant found with code TH1508221",
"timestamp": "2025-10-01 10:00:34 AM"
}
```

4.4.2 Verify Transaction Status

A request is initiated from the merchant platform, providing both the merchant code and the transaction ID. The transaction ID can either be the one generated by the merchant platform or the one generated by UrubutoPay. The system processes the request and returns the corresponding transaction status.

4.4.2.1 Request Method

HTTP request method	URI path segment
POST	/api/v2/payment/transaction/status

4.4.2.2 Request Header

Header name	Type
Content-Type	application/json
Authorization	api-key

4.4.2.3 Request Payload

Field Name	Type	Description	Required
merchant_code	string	Merchant code used during payment initiation	yes
transaction_id	string	Unique transaction of generated by merchant platform or generated by UrubutoPay	yes

4.4.2.4 Successful Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains payment details	yes
status	number	Http status code. Must be 200	yes
message	string	Successful confirmation message	Yes

4.4.2.5 Successful Data Object Response

Field Name	Type	Description	Required
payment_channel	string	Payment channels. **WALLET** or **CARD**	yes
payment_channel_name	string	**MOMO** or **AIRTEL_MONEY**	yes
transaction_status	string	**UrubutoPay Transaction statuses. We have INITIATED, PENDING, FAILED, VALID, PENDING_SETTLEMENT**	yes
internal_transaction_id	string	Unique transaction id generated by UrubutoPay	Yes
transaction_id	string	Unique transaction id generated by merchant platform	yes
merchant_code	string	Merchant code of merchant	yes
service_code	string	Service code of a merchant	yes
payer_code	string	The payer code or bill ID is used.	yes
amount	number	The amount paid	yes
payer_names	string	The names of the payer	yes
phone_number	string	The phone number used for payment	yes
currency	string	RWF / USD	yes
payment_date_time	string	Transaction date of a payment	yes
payer_email	string	Email of the payer	yes

4.4.2.6 Failure Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. YYYY-MM-DD HH:MM:SS	yes
status	number	Http status code. Any http code except 200, and 201	yes
message	string	Failure message	yes

Header's example:
Content-Type: application/json
Authorization: upg:test_sk_apikeywrwerwerwerwerwer

Request Payload:

```
{
"transaction_id": "45903423423432423438",
"merchant_code": "TH15082215"
}
```

Successful Response:

```
{
"timestamp": "2025-10-01 10:42:25",
"status": 200,
"message": "Successful",
"data": {
"transaction_status": "FAILED",
"payer_code": "201863794",
"payment_channel": "WALLET",
```

"payment_channel_name": "MOMO",

"currency": "RWF",

"merchant_code": "TH15082215",

"payer_names": "James Bond",

"payer_email": "janedoe@example.com",

"service_code": "breakfast-1818",

"payment_channel_transaction_ref_number": "23216116235",

"redirection_url": null,

"internal_transaction_id": "33202510011013041812",

"transaction_id": "45903423423432423438",

"payment_date_time": "01/10/2025 10:13:04",

"amount": 5,

"phone_number": "25078********"

}

}

Failure Response:

{

"timestamp": "2025-10-01 10:44:06 AM",

"status": 404,

"message": "payment not found"

}

4.4.3 Verify the ownership of the client's wallet account

This API allows merchants to verify if a customer is registered with **MTN MOMO** or **Airtel Money**.

Upon request, the merchant provides the payer's phone number, and the API responds with the registration status indicating whether the payer is enrolled with the specified mobile money provider.

4.4.3.1 Request Method

HTTP request method	URI path segment
POST	/api/v2/payment/account-holder/validate

4.4.3.2 Request Header

Header name	Type
Content-Type	application/json
Authorization	api-key

4.4.3.3 Request Payload

Field Name	Type	Description	Required
payment_channel_name	string	Wallet Payment channels. **MOMO** or **AIRTEL_MONEY**	yes
payer_phone_number	string	Phone number of the client.	yes

4.4.3.4 Successful Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which shows if customer is registered in MOMO or AIRTEL MONEY	yes
status	number	Http status code. Must be 200	yes

4.4.3.5 Successful Data Object Response

Field Name	Type	Description	Required
valid	boolean	Returns true or false.	yes

		true indicates that the client is registered with **MTN MOMO** or **AIRTEL MONEY** **false** indicates that the client does **not** have a wallet on **MTN MOMO** or **Airtel Money**	

Header's example:

Content-Type: application/json
Authorization: upg:test_sk_apikeywrwerwerwerwerwerwer

Request payload:

```
{
"payer_phone_number": "250786601003",
"payment_channel_name": "MOMO"
}
```

Successful Response:

```
{
"status": 200,
"data": {
"valid": true
},
"timestamp": "2025-10-01 10:47:11 AM"
}
```

Failed Response:

```
{
"statusCode": 400,
```

```
"error": "Bad Request",
"message": "Validation failed",
"validation": {
"body": {
"source": "body",
"keys": ["payment_channel_name"],
"message": "\"payment_channel_name\" must be one of [MOMO, AIRTEL_MONEY]"
}
}
}
```

4.4.4 Get full KYC for a client's wallet account

This API lets merchants retrieve a customer's full KYC details and confirm whether the customer is registered with **MTN MOMO** or **Airtel Money**.
The merchant submits the payer's phone number and payment channel. The API returns the registration status for the specified wallet provider along with any available KYC information.

4.4.4.1 Request Method

HTTP request method	URI path segment
POST	/api/v2/payment/account-holder/detail

4.4.4.2 Request Header

Header name	Type
Content-Type	application/json
Authorization	api-key

4.4.4.3 Request Payload

Field Name	Type	Description	Required
payment_channel_name	string	Wallet Payment channels. **MOMO** or **AIRTEL_MONEY**	yes
payer_phone_number	string	Phone number of the client.	yes

4.4.4.4 Successful Response

Field Name	Type	Description	Required
timestamp	date	Date time. Must have this format. **YYYY-MM-DD HH:MM:SS**	yes
data	object	Data object which contains customer details	yes
status	number	Http status code. Must be 200	yes

4.4.4.5 Successful Response DTO

Field Name	Type	Description	Required
given_name	string	First name of the customer/payer	yes
family_name	string	Family names of the payer	yes

Header's example:
Content-Type: application/json
Authorization: upg:test_sk_apikeywrwerwerwerwerwerwer

Request payload:

```
{
"payer_phone_number": "250786601003",
"payment_channel_name": "MOMO"
}
```

Successful Response:

```
{
"timestamp": "2025-10-01 11:37:20 AM",
"status": 200,
"data": {
"given_name": "Johdoe",
"family_name": "Rwanda",
"birthdate": "1997-08-06"
}
}
```

Failed Response:

```
{
"statusCode": 400,
"error": "Bad Request",
"message": "Validation failed",
"validation": {
"body": {
"source": "body",
"keys": ["payment_channel_name"],
"message": "\"payment_channel_name\" must be one of [MOMO, AIRTEL_MONEY]"
}
}
}
```

5. Transfer Guides

UrubutoPay allows merchants to send transfers to their beneficiaries, either as a single transfer or in bulk. Transfers can be funded through the UrubutoPay wallet or other supported payment channels.

The transfer can be done using Transfer APIs or Through Web Interface provided by UrubutoPay

5.1 Overview

UrubutoPay currently supports the following transfer channels:

5.1.1 UrubutoPay Wallet to MTN MOMO

Merchants must first load funds into their UrubutoPay wallet. They can then send money to their beneficiaries through the Portal or via APIs. Beneficiaries must have an MTN phone number.

5.1.2 UrubutoPay Wallet to Airtel Money

Merchants must first load funds into their UrubutoPay wallet. They can then send money to their beneficiaries through the Portal or via APIs. Beneficiaries must have an Airtel phone number.

5.1.3 MTN MOMO to MTN MOMO

We use merchant MTN MOMO Wallet and send funds to Their beneficiaries. Beneficiaries must have an MTN phone number.

5.1.4 Airtel Money to Airtel Money

We use merchant Airtel Money Wallet and send funds to Their beneficiaries. Beneficiaries must have an Airtel phone number.

5.2 Transfer APIs

Transfer APIs enable merchants to securely initiate and manage transfers directly from their own platforms, allowing seamless integration of payment capabilities into their systems.

5.2.1 Initiate Transfer

The initiation of API allows merchants to submit transfer requests to UrubutoPay. Each request

includes the selected payment channel and beneficiary details.

This API is secured with strict validation measures. Before sending a request, merchants must ensure that their public IP addresses are whitelisted and that their API key is valid. If any of these requirements are not met, the system will reject the request.

5.2.1.1 Method

HTTP request method	URI path segment
POST	/v2/bulk-disbursement/initiate

5.2.1.2 Header

Header name	Type
Content-Type	application/json
Authorization	upg:test_sk_sdufsdfsdfmvsdfsdfsdfsdfsd

5.2.1.3 Parameters

Field Name	Type	Description	Required
source_payment_channel	string	**MERCHANT_WALLET, MOMO,** and **AIRTEL_MONEY**	yes
destination_payment_channel	string	**MTN** and **AIRTEL_MONEY**	yes
source_of_funds	string	If the source payment channel is MTN or AIRTEL_MONEY, the source of funds must come from a Rwandan phone number; otherwise, the transaction cannot be processed.	yes
beneficiary_to_be_charged	string	Yes or No. If set to **Yes**, your clients will be charged. If set to **No**, you will cover the charges on behalf of your clients.	yes
external_transaction_ref_number	string	Unique transaction id generated by your system	yes
confirmation_email	string	If the source payment channel is MTN or	yes

		Airtel Money, a confirmation email will be provided. This allows us to send the confirmation details to the email address supplied.	
beneficiaries	Array of object	Beneficiary details	yes

5.2.1.3.1 Beneficiaries

Field Name	Type	Description	Required
beneficiary_names	string	Names of beneficiary	yes
beneficiary_phone_number	string	If the source payment channel is **MTN**, the phone number must be an MTN number. Likewise, if the source payment channel is **AIRTEL_MONEY**, the phone number must be an Airtel number.	yes
amount	number	Amount to be transferred	yes
description	string	Description	no
beneficiary_code	string	Unique beneficiary code generated by your system. By providing this code will enable you to verify specific transfer of beneficiary	yes
currency	string	Currency must be **RWF**	no

5.2.2 Success Response

Field Name	Type	Description	Required
status	number	200, Http Status code	yes
message	string	Success message	yes
data	object	Object containing disbursement details	yes

5.2.2.1 Data

Field Name	Type	Description	Required
source_payment_channel	string	Payment channel. MTN, AIRTEL_MONEY & MERCHANT_WALLET	yes
destination_payment_channel	string	MTN or AIRTEL_MONEY	yes
amount	number	Total amount	yes
transaction_date	string	Transaction Date	yes
transaction_status	string	By default, if the source payment channel is **MERCHANT_WALLET**, the transfer status will be marked as **SUCCESSFUL**. For all other channels, the status will initially be **PENDING** because we must first debit the funds from your MoMo wallet before transferring them to your beneficiary.	yes
disbursement_status	string	Status determines whether disbursement is Success, Partial Success, or Failed. The statuses are the following:	yes

		FULL_SUCCESS, PARTIAL_SUCCESS and **FAILED**	
currency	string	Currency. RWF	yes
confirmation_status	string	Must be **APPROVED, EXPIRED** or **ACTIVE**	yes
external_transaction_ref_number	string	Transaction Id provided by your system	yes
internal_transaction_ref_number	string	Transaction Id generated by UrubutoPay	yes

5.2.3 Failure Response

Field Name	Type	Description	Required
status	number	Http Status code	yes
message	string	Success message	yes